

URL Shortener Planning

Purpose of the System :-

- Create a basic URL Shortening Service using C++.
- The application should allow users to enter a long URL, generate a shortened URL, and **retrieve the original URL from the shortened version.**

How It Works :-

1. The user enters either a **long URL** or a **short URL**
2. The system validates the entered URL.
 - **If the URL is invalid** : An error message such as **"Invalid URL"** is displayed
 - **If the URL is valid** : The process continues.
3. If the input is a **long URL**:
 - The system checks the file to determine whether the same long URL already exists.
 - **If it exists** : The previously generated short URL associated with it is returned.
 - **If it does not exist** :
 - The system generates a new short URL
 - Stores the mapping between the long URL and short URL in the file
 - Then returns the new short URL.
4. If the input is a **short URL**:
 - The system searches the file for the matching short URL.
 - **If found** : the corresponding original long URL is returned.
 - **If not found** : an error message such as **"Short URL not found"** is displayed.

Requirements:-

Phase 1 : URLValidatorModule

- **Responsible for** : validating user input URLs.

Function:

- `bool IsValid(string Url)`
 - It receives a URL from the main program and checks if it is valid.
 - Must check
 - starts with http:// or https://
 - no empty string
 - no spaces
 - contains domain name (ex : .com)
 - Returns `true` if the URL is valid, otherwise returns `false`
 - `bool HasValidScheme(string Url )`
 - Checks whether URL starts with:
 - http:// and https://
 - `bool HasNoSpaces(string Url )`
 - Ensures URL has no spaces.
 - `bool HasValidDomain(string Url)`
 - Checks domain exists like:
 - google.com
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Phase 2 : ShortCodeGenerator

- **Responsible for** : generating a unique and human-readable short code.

Function:

- `string Generate(string LongUrl)`
 - It only generates the code and used by URLShortenerService
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Phase 3: File Management Module

- **Responsible For :** Handle saving and loading data from file.

Functions:

- `bool FileExists()`
→ Checks if the storage file exists.
 - `void CreateFile()`
→ if file doesn't exist create new one
 - `void SaveMapping(string LongUrl, string ShortUrl)`
→ Saves a new URL mapping into the file.
 - `string ReturnLongURL(string ShortUrl)`
→ Finds and returns the original URL.
 - `string ReturnShortURL(string LongUrl)`
→ Finds existing short URL for a long URL.
 - `bool ShortCodeExists(string Code)`
→ Checks if short code already used to help URLShortenerService
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Phase 4: URLShortenerService

- **Responsible for :** Main logic controller of the system

Functions:

- `string ShortenURL(string LongUrl)`
 - Validate URL from Validator class
 - Check if already exists
 - If exists → return old short URL
 - Else generate new code
 - Save mapping
 - Return short URL

- `string ResolveURL(string ShortUrl)`
→ Returns the original long URL from a short URL.
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Phase 5: Controller

- **Responsible for :** Shows menu displays results, and repeats until exit.

Functions:

- `void Run()`
→ Starts the whole system.
 - `void ShowMenu()`
→ Displays:
 1. Shorten URL
 2. Resolve URL
 3. Exit
 - `void HandleShorten();`
→ Gets long URL from user and shortens it.
 - `void HandleResolve()`
→ Gets short code and resolves it.
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